

Secure Hospital Network Project

This project is a complete VLAN-based hospital network designed in Cisco Packet Tracer. The network includes VLAN segmentation, inter-VLAN routing, wireless access, security using ACLs, cameras, and server connectivity.

Project Devices

- Cisco 2911 Router - 3 Cisco 2960 Switches - Access Point - 1 Server - 6 PCs - 2 Cameras - Copper Straight-Through Cables

VLAN Structure

VLAN 10 - ADMIN - 192.168.10.0/24
VLAN 20 - DOCTORS - 192.168.20.0/24
VLAN 30 - LAB - 192.168.30.0/24
VLAN 40 - CCTV - 192.168.40.0/24
VLAN 50 - WIFI - 192.168.50.0/24

Router Configuration

```
interface g0/0  
no shutdown
```

```
interface g0/0.10  
encapsulation dot1Q 10  
ip address 192.168.10.1 255.255.255.0
```

```
interface g0/0.20  
encapsulation dot1Q 20  
ip address 192.168.20.1 255.255.255.0
```

```
interface g0/0.30  
encapsulation dot1Q 30  
ip address 192.168.30.1 255.255.255.0
```

```
interface g0/0.40  
encapsulation dot1Q 40  
ip address 192.168.40.1 255.255.255.0
```

```
interface g0/0.50  
encapsulation dot1Q 50  
ip address 192.168.50.1 255.255.255.0
```

Switch Configuration

Create VLANs on all switches:

```
vlan 10  
name ADMIN
```

```
vlan 20  
name DOCTORS
```

```
vlan 30  
name LAB
```

```
vlan 40  
name CCTV
```

```
vlan 50  
name WIFI
```

Trunk Configuration

```
interface fa0/24  
switchport mode trunk
```

Access Port Configuration

Example for VLAN 10:

```
interface fa0/1  
switchport mode access  
switchport access vlan 10
```

ACL Security

```
access-list 100 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255  
access-list 100 permit ip any any
```

```
interface g0/0.30  
ip access-group 100 in
```

Wireless Configuration

```
SSID: Hospital_WiFi  
Password: hospital123  
Access Point connected to VLAN 50.
```

Common Problems and Solutions

1. Red Link Problem:
Cause: Interface shutdown or no power adapter.

Solution: Use 'no shutdown' or add power adapter to Access Point.

2. APIPA Address 169.254.x.x:

Cause: DHCP failed.

Solution: Configure static IP manually.

3. Request Timed Out Between VLANs:

Cause: Trunk not configured or router subinterfaces missing.

Solution: Configure trunk and Router-on-a-Stick correctly.

4. Access Point Not Working:

Cause: No power adapter installed.

Solution: Add Access Point Power Adapter.

5. Devices Cannot Ping:

Cause: Wrong VLAN assignment or gateway configuration.

Solution: Verify VLANs and default gateways.

Testing

- Ping devices in same VLAN.
- Ping between VLANs.
- Test WiFi connectivity.
- Verify ACL blocking.
- Use 'show vlan brief' and 'show interfaces trunk'.